Yashvanth S

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EDUCATION

IIITDM Kancheepuram – Integrated Dual Degree (B.Tech + M.Tech) in CSE **CGPA: 9.32/10 (Sem 4)** | Expected 2029
Highest CGPA in CSE cohort | *Relevant:* DSA, OOP, DBMS, Design & Analysis of Algorithms

Velammal Matriculation Hr. Sec. School, Chennai – Higher Secondary (Class XII) **97.33%** | 2024

EXPERIENCE

Artificial Intelligence Intern | *Random Walk*, Chennai Jun 2026 – Aug 2026
TypeScript, Three.js, MediaPipe, Gaussian Splatting, FastAPI, Modal, Docker | doppel-rw.vercel.app

- Built **Project Doppler**, a head-tracked display that turns a monitor into a window: **MediaPipe Face Landmarker** in a Web Worker → **One-Euro filter** → off-axis (Kooima) projection re-rendering a **Gaussian-splat** scene from the viewer's eye position.
- Ported the generalized off-axis perspective maths to TypeScript with parity tests against the Python reference; replaced exponential smoothing with a One-Euro filter to cut jitter without adding lag.
- Shipped the photo-to-3D pipeline: **FastAPI** gateway in front of an **Apple SHARP** worker on a **Modal T4 GPU**, with content-addressed caching, in-flight request coalescing, job restore after restart and API-key auth.

Founding Engineer | *Promptli*, Toronto (remote, part-time) Feb 2026 – Sep 2026
LiveKit, OpenAI Realtime, Twilio SIP, Convex, Cal.com, AWS Lightsail, Docker | promptli.com

- Built an end-to-end voice-AI receptionist on **LiveKit Agents** (TypeScript) with the **OpenAI Realtime API** for low-latency speech-to-speech, handling inbound and outbound PSTN calls over Twilio SIP trunks.
- Engineered **14 LLM function-calling tools** (availability, booking, reschedule, RAG knowledge search, customer history, live escalation) wired into self-hosted Cal.com for autonomous 24/7 appointment management.
- Shipped **4 outbound agent modes** with per-tenant prompt templates; pre-hydrated customer context to cut tool-call latency.
- Designed the real-time data layer on **Convex** (21-table schema, vector search for RAG) and automated per-tenant infra with cloud-init **AWS Lightsail** VMs running Docker Compose stacks.

Software Developer Intern | *Artly Creative Labs* (ReelCat) Jan 2026 – Feb 2026
React, TypeScript, Remotion, AWS Lambda, Convex, React Native (Expo)

- Built **ReelCat**, a video-editing SaaS with timeline editing, overlays and real-time preview, plus an **Expo** mobile app.
- Architected a serverless render pipeline on **Remotion + AWS Lambda**, cutting render time by **60%** vs client-side.
- Integrated auth, payments (Dodo), analytics (PostHog) and real-time sync (Convex).

Lead Full-Stack Developer | *E-Summit '26, IIITDM Kancheepuram* Jan 2026 – Feb 2026
Next.js, TypeScript, PostgreSQL, Supabase, Redis, Vercel | github.com/YashvanthSankar/esummit26

- Solely built the official E-Summit '26 platform (**194 commits**): **12,000+** page views, **1,000+** concurrent users, zero downtime.
- QR ticket verification with **Redis caching** at **sub-100 ms** for **1,000+ attendees**; admin dashboard with revenue analytics, CSV exports, Row Level Security, Google OAuth and PWA support.

PROJECTS

choco.video – On-device long-video-to-clips app for macOS (open source) *Rust, Tauri 2, Svelte 5, whisper.cpp, FFmpeg*

- Rust pipeline: transcribe → sentence-bounded windows → **BM25 + audio-quality prior** shortlist → batched LLM scoring with retry and context-halving → **YuNet face tracking** and active-speaker reframing → captioned 9:16 render; a 47:52 interview yields 4 clips in 226 s on an M3 Pro.
- Shortlist defaults chosen by measured ablation (**93%** of golden clips reachable vs 47%); hardware-encoder failover mid-clip (VideoToolbox/NVENC/AMF); CI on macOS and Windows with snapshot tests, clippy and cargo-deny; signed DMG with auto-update.

GudTour – Multilingual disaster alerts for tourists, Android + iOS *Expo, React Native, Convex, TypeScript*

- Native Android and iOS apps from one Expo codebase; ingests official **NDMA CAP** feeds with hardened XML parsing, geofenced delivery via geohash fan-out with idempotent checkpointed jobs, six languages, **308 automated tests**.
- **Placed at CS Week 2026**, IIITDM Kancheepuram. gudtour.zapdos.me

Sagar Prahari – Govt. of India coastal-campaign app for MoES / NIOT *Expo, Convex, TypeScript, React Native*

- 7-day campaign app **deployed by Govt. of India on the Play Store**: 7 events incl. the CISF Cyclothon (**4.2 Cr+ participants**), OTP login, registration, nearest-beach suggestions, photo + location attendance, points and streaks.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Rust, Swift, SQL, C++
Frontend: React, Next.js, React Native (Expo), Svelte, Tauri, Tailwind CSS, Three.js / WebGL
Backend & AI: Node.js, Express, FastAPI, Convex, LiveKit Agents, OpenAI / Groq / Gemini, RAG, whisper.cpp, Zod, BullMQ, WebSockets
Data: PostgreSQL, Supabase, Prisma, MongoDB, Redis, Convex
Infra & Tools: Git, Docker, AWS (EC2, Lambda, Lightsail), Vercel, Cloudflare, Modal, nginx, FFmpeg, Playwright, Linux

ACHIEVEMENTS & LEADERSHIP

- **Solely conducted** E-Summit '26 Hackathon & Ideathon – **600 participants** from **50+ colleges**, highest external turnout in event history
- **2nd Place** (RetailTech/SaaS) and **Top 28** of **750+ teams** at VOID:V1 among **3,000+** participants | **Top 100** of **2,000+ teams** at Sprintathon '25
- **Highest CGPA** in second-year CSE cohort | **97.95 percentile** in JEE Main 2024 | **1st Place** in **3 hackathons** at CS/Developers Club, IIITDMK

CERTIFICATIONS

Supervised Machine Learning (*deeplearning.ai*) | Web Developer Bootcamp (*Udemy*) | Python & ML (*Kaggle*) | SQL (*HackerRank*)